Reply to reports on article JBES-P-2022-0567.R1

First of all, we would like to once more thank the editor, the associate editor and the reviewers for their thoughtful comments and suggestions which have greatly helped to improve the paper. In the revision, we have addressed all points raised. Please see our point-by-point responses below.

Reply to the associate editor

1. *(a) My apologies, but I maintain that your H_0 (page 3) is unnatural and uninteresting. In the trend analysis of multiple time series, it is extremely unlikely that the natural 'position of ignorance' is that all trend curves are the same. What if, for example, I already know or suspect that the curves form distinct groups? It is then something of a wasted effort to start from the position that they do not differ at all.*
   (b) Also, the null hypothesis is not robust to high-dimensionality as it is more likely that some curves will differ as I observe a higher number of variates (I understand that you do not do this in the paper but this only makes it even less interesting, in my view).
   (c) As a cartoon illustration of the first point above, if I suspect that $p-1$ (say) curves may be the same, but 1 is distinctly different, do I need to first remove the offending one before performing your test, in which case I enter the territory of post-selection inference (which you do not cover), or do I not need to remove the offending one, in which case I am pretending that my H_0 is different from what it is in reality?

   (a) We fully agree with you that – as you put it – our global null hypothesis H_0 ("all trends are exactly the same") is unnatural and uninteresting. In most applications, it is rather obvious that not all trends are the same. Hence, there is no need to formally test the hypothesis H_0 just to get the unsurprising answer that it is violated. However, it is very interesting and practically relevant (at least in our point of view) to understand in which ways H_0 is violated: which trends are different, in which time periods the trends differ, whether there are groups of (more or less) identical trends, etc. The main contribution of our paper is to develop statistical methods which provide this very relevant information. Specifically, we develop

   (i) methods to make simultaneous confidence statements about which trends are different and where (i.e., in which time intervals) they differ from each other

(ii) clustering methods for finding groups of time trends (which come with certain statistical confidence guarantees).

We develop these methods in the context of a classical statistical testing framework. In this framework, the global null hypothesis H_0 appears as an auxiliary construct/concept which, however, is not interesting by itself but only a means to formally write up the methods/theory.
In summary, your criticism doubtlessly applies to standard tests of H_0 which only allow to say (with a certain statistical confidence) whether H_0 is violated or not, and which are thus very uninformative. However, it does not apply to our approach. Rather, our approach can be regarded as – at least a partial – answer to your criticism of standard uninformative tests of H_0. In the introduction of the paper, we have emphasized this crucial difference between our approach and standard tests of H_0 as clearly as possible.

(b) This is a fair point. A high-dimensional extension would indeed be very interesting. However, adapting the theory would be highly non-trivial, which is why we have not tackled this extension in the paper.

(c) Simply run our methods on all p curves. Then (if your guess is correct, i.e., if all curves are the same except for one), our methods will give you the following information (provided that the regularity conditions of our theory are satisfied):

- (Asymptotically with probability $1 - \alpha$) our inference methods will tell you (i) that all trends are the same except for one and (ii) in which time intervals this trend differs from the others.
- (Again asymptotically with probability $1 - \alpha$), the clustering algorithm will give you two groups: a cluster with the $p - 1$ identical trends and a singleton cluster with the other trend.

Note that in the simulation study, we have used precisely such a cartoon setting with $p-1$ identical trends and 1 different trend to showcase the power properties of our inference methods.

2. *The clustering part is nice and perhaps it could form the main thread of a much shorter paper.*

We are happy to hear that you find at least the clustering part nice. However, for the reasons laid out in our response to your previous comment, we regard

(i) our inference method to make simultaneous confidence statements
(ii) our clustering algorithm

as the two main, equally important contributions of the paper, which provide complementing information in practice. We would thus prefer not to drop (i) from the paper. We hope you can understand.

3. *The empirical application is interesting, but this is thanks to the use of clustering rather than as a consequence of testing - the null is obviously inappropriate, as Figure 6 suggests, so there is no point in assuming it only to surely reject it. In other words, "let us test our null of equality between all trends" doesn't seem a scientifically valid suggestion for this dataset. The suggestion of clustering the countries is fine, though.*

   Once again, we appreciate that you find the clustering part of the empirical application interesting. However, we strongly disagree that the other results (based on our simultaneous inference method) are uninteresting. In our view, they add further relevant aspects to the clustering part. Among other things, they provide information on the time periods in which the trends differ:

   - We detect differences between the trends of Australia and the Netherlands / of Belgium and the Netherlands starting only from 1968 /1966 onwards. This is in line with Knoll et al. (2017) who report that before World War II the countries exhibit similar trends, while the trends start to diverge sometime after World War II.
   - Contrary to Knoll et al. (2017), we also find significant differences before World War II, specifically, between the time trends of Belgium and Denmark / of Belgium and the USA.

   This illustrates why we regard our simultaneous inference and clustering methods as equally important tools that complement each other.

4. *Can your model (1.1) include autoregression? Either way, it would be good to see it spelled out explicitly.*

   Model (1.1), which is given by the equation

$$Y_{it} = m_i\Big(\frac{t}{T}\Big) + \boldsymbol{\beta}_i^\top \boldsymbol{X}_{it} + \alpha_i + \varepsilon_{it},$$

   does not include autoregression because we have assumed that the regressors $\boldsymbol{X}_{it}$ are stationary. In the autoregressive case

$$Y_{it} = m_i\Big(\frac{t}{T}\Big) + \sum_{j=1}^{d} \beta_{i,j} Y_{i,t-j} + \alpha_i + \varepsilon_{it}$$

   with $\boldsymbol{X}_{it} = (Y_{i,t-1}, \dots, Y_{i,t-d})^\top$, this stationarity assumption is violated in general (due to the inclusion of the trend function m_i in the model). It should be possible though (although not straightforward) to extend our theory to locally stationary covariates, which would include the autoregressive case. We have spelt this out explicitly on p.8 in the revised paper.

5. *I was unconvinced by the subtraction of the overall level on p. 6.*
   (a) Firstly, this clearly risks eliminating useful information (what if some curves differ from others in the overall level)?
   (b) Secondly, I am not sure if it is meaningful to look at averages of (e.g. increasing) trends - for example, what is the interpretation of a "long-term average house price level in [a] country"? In "Time Series 101", we teach students not to take averages of non-stationary time series!

   (a) Due to the inclusion of the fixed effect α_i, the trend function m_i in our model (eq. 1.1 in the paper) is only identified up to the overall level. More specifically, as described on p.6 of the paper, our model

$$Y_{it} = m_i\Big(\frac{t}{T}\Big) + \boldsymbol{\beta}_i^\top \boldsymbol{X}_{it} + \alpha_i + \varepsilon_{it} \tag{1}$$

   is observationally equivalent to

$$Y_{it} = m_i^\diamond\Big(\frac{t}{T}\Big) + \boldsymbol{\beta}_i^\top \boldsymbol{X}_{it} + \alpha_i^\diamond + \varepsilon_{it}, \tag{2}$$

   where $m_i^\diamond = m_i + c_i^\diamond$, $\alpha_i^\diamond = \alpha_i - c_i^\diamond$ and $c_i^\diamond$ is an arbitrary real constant. Put differently, the observed data $\{(Y_{it}, X_{it})\}$ are consistent with the trend m_i (and the fixed effect α_i) as well as with the vertically shifted version $m_i^\diamond$ (and $\alpha_i^\diamond$). Hence, the data do not contain any information on the overall level of the curve m_i. Therefore, we do not eliminate this information by subtracting the overall level for identification.
   Importantly, this has the following consequence: It is not possible to test whether the curves m_i have the same overall level. We can only test whether the curves are parallel, i.e., whether they are identical up to the overall level. If all curves are normalized to have the same overall level 0 by our identification strategy, then this parallelism hypothesis coincides with the hypothesis that the trends are identical (which is our global null hypothesis H_0).

   (b) As far as we can see, normalizing (trending) time series by subtracting long-term averages is quite common in practice. In climatology, for instance, people usually analyze temperature anomalies (i.e., the deviation of the temperatures from a long-term average) rather than the raw temperatures themselves. We do the same in the context of (log) house prices for the following reason: Suppose we want to compare house prices (say) in London to the prices in a smaller city like Derby, UK. Obviously, the house prices in London have been way above those in Derby in the last decades. Thus, the time trend of the London prices will lie way above the Derby trend, implying that the two trends are different everywhere. The interesting question is whether the time trends are parallel, i.e., whether the prices have evolved "in the same way" over time. To test this parallelism hypothesis (and infer in which ways it is violated), we normalize the trend curves to have overall level 0, which makes parallel curves identical (as discussed in the last paragraph of our response to part (a) above).

We have revised the text in Section 2.2, p.6/7 of the paper to make the above points clearer.

6. *(C5) - you use the delta notation before defining it. Why do your covariates X have to be random, e.g. why can't you have one of them as (say) a linear trend?*

   Thank you for pointing out the undefined use of the delta notation in (C5). We have fixed this.

   Regarding your second point on the randomness of the covariates: The problem is that the trend function m_i in our model (eq. 1.1 in the paper) is not identifiable when a covariate is completely deterministic (because in this case, one cannot distinguish whether the time trend comes from the function m_i or from the regressor component). To see this, consider the following example: suppose we only have one covariate X_{it} in the model and assume that X_{it} is a linear trend i.e., $X_{it} = t/T$. In this case, our model reads

$$Y_{it} = m_i\Big(\frac{t}{T}\Big) + \beta_i \cdot \frac{t}{T} + \alpha_i + \varepsilon_{it}.$$

   Here, the trend function m_i is not identified because

$$m_i\Big(\frac{t}{T}\Big) + \beta_i \cdot \frac{t}{T} = \Big\{m_i\Big(\frac{t}{T}\Big) + \beta_i' \cdot \frac{t}{T}\Big\} + \big\{\beta_i - \beta_i'\big\} \cdot \frac{t}{T}$$

   for any $\beta_i' \in \mathbb{R}$. Hence, we can freely shift a linear trend (with arbitrary slope parameter β_i') between the trend function and the regressor component of the model.
   Please note that the case of fully deterministic covariates is similar to the case of strongly dependent random regressors discussed in Remark 2.1. There we point out that it is difficult to handle strongly dependent random covariates because they tend to produce stochastic trends that are difficult to discern from the deterministic trend m_i. If the trend produced by the covariates is not stochastic but deterministic (as in the example discussed above), then it is not only difficult to discern this trend from the function m_i but simply impossible (i.e., m_i is not identifiable).

7. *Your comment (iii) on p. 9. But isn't it the case that for near-unit root processes, you will have a variety of different scenarios (some stochastic trends pointing strongly upwards, some downwards, some with no specific direction) - a situation that should be easy to discriminate from your typical scenarios of interest?*

   We are not fully sure whether we get your point. – Importantly, we only observe one sample path $\mathcal{Y}_i = (Y_{i1}, \dots, Y_{iT})$ per time series i (neglecting covariates) over a fixed amount of time (T fixed) in practice. Suppose we plot the observed path $\mathcal{Y}_i$ for some i and we see a strong upward trend in the data. Such a strong upward trend may be produced by a deterministic trend function or by an AR process with near-unit root. If we had the luxury of observing longer and longer sample paths (i.e., $T \to \infty$), it would get easier and easier to figure out which trend model is more appropriate. E.g., if the observed time series path does not stop trending upwards as T gets larger and larger, then we probably do not have a stationary near-unit root AR process in the background (as we have to revert to the mean in such a stationary model after a while). However, for fixed moderately large T, it is fairly difficult to reliably distinguish between a deterministic trend and a stochastic trend produced by a near-unit root AR process. This is essentially all that we wanted to say in Remark 2.1 (iii) on p.9.

8. *P. 11. "Throughout the paper, we assume that the long-run error variance is the same for all i, (...) [but we] use a different estimator of σ^2 for each i - this seems suboptimal.*

   Yes, this is suboptimal but avoids certain theoretical complications. We now mention this on p.12.

9. *I was disappointed to see that the method was not able to handle missing values.*

   If data points are "missing at random" and not too many data points are missing, then it is actually straightforward to adjust our methods. In general, however, it is non-trivial to do so. We comment on this very briefly in Section 8.

Reply to referee 1

1. *To address some "big picture" concerns about practical relevance, the authors clarify that rather than the null hypothesis of "no differences ever", the main focus is on the alternative hypotheses, and specifically the ability to detect 1) which series are different and 2) when. This is a reasonable response. However, the simulation studies do not really showcase these priorities.*
   (a) First, simulation results do not appear at all in the main paper. If there is a question about practical relevance of the methods (which the review team did raise), then empirical support should be elevated above some of the technical details.
   (b) Second, the global size and power tables do not clearly illustrate such local testing capabilities—these metrics would require a much larger collection of data-generating processes to measure performance across a large spectrum of alternative hypotheses.
   (c) Third, the suggested competitors have not been included, and no compelling

reasons were given (more below).
*(d) Finally, the authors have kept $m_i = 0$ to simulate data under the null "WLOG". As I noted before, this is not WLOG for all methods; that it *is* WLOG for their method suggests that they may observe comparative gains over other approaches for smaller sample sizes (n).*

(a) We fully agree with you. However, we simply do not know how to incorporate (part of) the simulations given the 35-page-limit. Even if we dropped all theoretical results from the paper (i.e., Section 4 and part of Section 5, around 5 pages in total) and added only simulation parts A.1 (size and power of the multiscale test) and A.4 (performance of the clustering algorithm) (around 10 pages in total) to the paper, we'd be far from the 35-page-limit. To give at least a bit more weight to the simulations, we have added a short paragraph to the revised paper (see the new Section 6) which provides an overview of the simulation study.

(b) In the simulation study, we compute the global power (and the global size) as the percentage of simulation runs in which the test rejects the global null H_0 ("all trends are the same"). We agree that this quantity is not very suitable to capture local testing capabilities. In the revision, we compute further metrics – specifically, further types of power – which we think are more apt to capture such local testing capabilities.
We briefly motivate the considered metrics. Suppose our test rejects the local null hypothesis $H_0^{[i,j]}(u,h)$ under which the trends m_i and m_j are identical on the interval $[u-h, u+h]$. We say that the test finds

- a **real difference** between i and j on $[u-h, u+h]$ if $H_0^{[i,j]}(u,h)$ is indeed violated (i.e., if the trends m_i and m_j are indeed different on $[u-h, u+h]$)
- a **spurious difference** between i and j on $[u-h, u+h]$ if $H_0^{[i,j]}(u,h)$ is not violated (i.e., if the trends m_i and m_j are the same on $[u-h, u+h]$).

Notably, the concept of global power does not give any information about

- whether the differences found by the test are real or spurious
- which real differences between the trends have been found, in particular, whether all or only part of the real differences have been found.

We refine the concept of global power to give this kind of information. More specifically, we compute the following types of power in our simulation setup where the trends $m_2, \ldots, m_n$ are identical and m_1 differs from them only in the time period $\mathcal{I}^* = [0.2, 0.4] \cup [0.6, 0.8]$:

- **real power**$^{[1]}$: the percentage of simulation runs in which our test finds a real difference between m_1 and *at least one* other trend m_i [i.e., a difference between m_1 and another m_i on an interval $[u-h, u+h]$ with $[u-h, u+h] \cap \mathcal{I}^* \neq \emptyset$].
- **real power**$^{[50]}$: the percentage of simulation runs in which our test finds a real difference between m_1 and *at least 50%* of the other trends m_i [i.e., a difference between m_1 and m_i on an interval $[u-h, u+h]$ with $[u-h, u+h] \cap \mathcal{I}^* \neq \emptyset$ for at least 50% of the other trends i].
- **real power**$^{[100]}$: the percentage of simulation runs in which our test finds a real difference between m_1 and *all* other trends m_i [i.e., a difference between m_1 and m_i on some interval $[u-h, u+h]$ with $[u-h, u+h] \cap \mathcal{I}^* \neq \emptyset$ for all $i \neq 1$].
- **spurious power**: the percentage of simulation runs in which our test finds at least one spurious difference.

We report the performance of our test in terms of the real and spurious power numbers defined above in Section A.1 of the revised simulation study.

(c) Please see our reply to your next comment 2 for the details on the competitors we have included in the revision.

(d) In the revision, we simulate data under the null

$$H_0 : m_i = m \text{ for all } i$$

not only with $m \equiv 0$ but also with the bump function

$$m(u) = b \cdot \vartheta(u, 0.3, 0.1) - b \cdot \vartheta(u, 0.7, 0.1),$$

where $\vartheta(u, c, d) = \mathbb{1}\{\frac{|u-c|}{d} \leq 1\}(1 - \frac{(u-c)^2}{d^2})^2$ and $b = 0.25$ (see Figure A.1 for a plot of this function). We were reluctant to include size simulations with different choices of m because our theory suggests that the specific form of m should not matter. However, the theory is only asymptotic and thus ignores certain small sample effects that are discussed below. Hence, the specific form of m is indeed relevant to some extent. Thank you for remaining persistent on this point (which we have not realized before). We apologize for not having included size simulations with different choices of m earlier.
Table 1 shows the empirical size numbers for the two considered choices of m (i.e., m the zero / bump function). As can be seen: (i) When m is the bump function, the test is conservative. In particular, the size numbers are substantially below the target α for the smallest sample size $T = 100$, gradually getting better as the sample size increases. (ii) When m is the zero function, in contrast, the size

(a) zero function

	nominal size α		
T	0.01	0.05	0.1
100	0.006	0.035	0.070
250	0.020	0.079	0.134
500	0.012	0.058	0.112

(b) bump function

	nominal size α		
T	0.01	0.05	0.1
100	0.001	0.005	0.014
250	0.006	0.031	0.068
500	0.011	0.049	0.093

Table 1: Empirical size numbers with m the zero and the bump function.

numbers tend to be rather on the liberal than the conservative side. (iii) For smaller sample sizes, in particular for $T = 100$, the size numbers tend to be more accurate when m is the zero function. For larger sample sizes, in contrast, in particular for $T = 500$, the size numbers are fairly accurate no matter whether m is the zero or the bump function. How can these differences in size numbers be explained? Importantly, estimating the long-run variance σ^2 of the error terms ε_{it} is more difficult in the presence of a non-trivial (i.e., non-constant) trend m. In particular, if we do not manage to eliminate the trend m accurately from the data (e.g. by differencing or nonparametric regression techniques), this will bias the estimator of the long-run error variance. This is illustrated in Figure 1: whereas the estimates of the long-run error variance are nearly unbiased in the case with $m = 0$, there is a clear bias in the case where m is the bump function (the bias getting smaller as the sample size increases). As the long-run error variance enters our multiscale test statistic $\widehat{\Psi}_{n,T}$ as a normalization constant, a biased estimate of it will in general lead to size distortions.

2. *I do not understand why the authors are resistant to add the simple competitors suggested (additive models w/ CI-based testing and Bayesian analogs). Their reasoning was 1) these methods do not offer the same theory guarantees and 2) they include another method, sizer, and a simplified version of their approach. For 1): if the theory actually offers practical gains, then this should show up clearly empirically and further advertise the proposed methods! For 2): sizer is a reasonable competitor (for n=2), but I don't see why that would eliminate consideration of any alternatives. If the question is "which curves are different and when", a natural first approach would be to fit a nonparametric model to each curve and see where/when the confidence/credible intervals do not overlap (w/ or w/o a Bonferroni correction). Beating this approach will be more convincing for general readers.*

   Following your suggestion, we have added a simple competitor which is based on uniform confidence bands. A detailed description of the competitor and the results of our comparison study can be found in Section A.3 of the supplementary

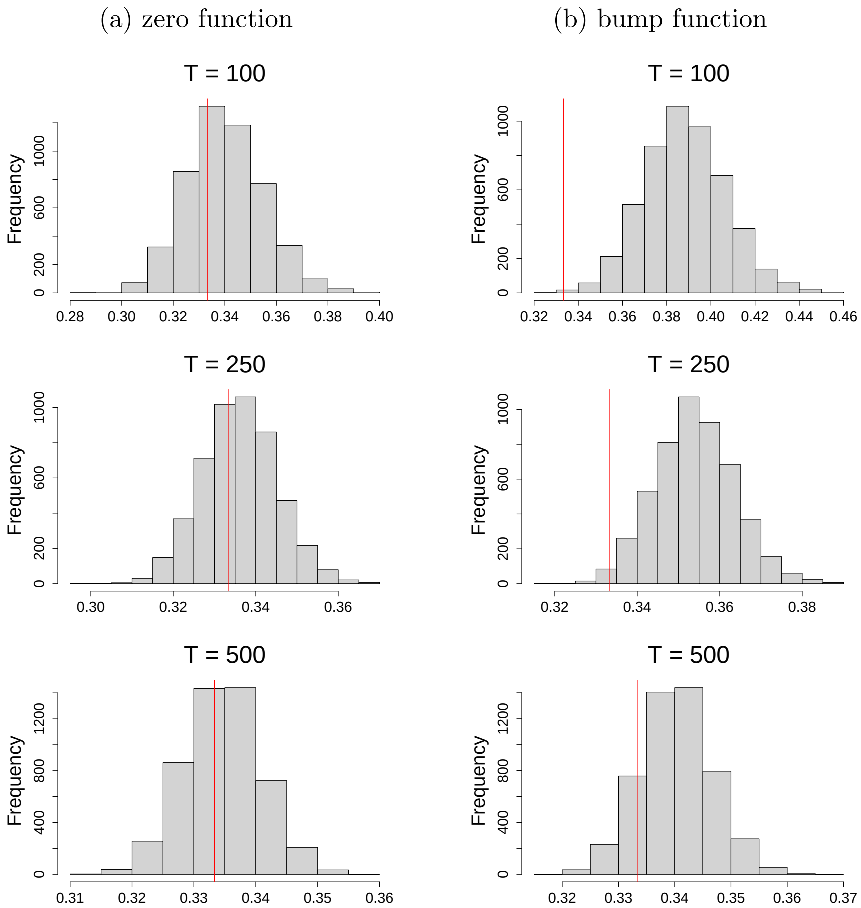

Figure 1: Histograms of the estimated (square root of the) long-run error variance with m the zero and the bump function. The red vertical line is the true value of the (square root of the) long-run error variance.

material. The simulation results reported there show that our approach vastly outperforms the competitor.

We have also considered the proposed Bayesian competitor from Meyer et al. (2015) and Zemplenyi et al. (2021)[1] which uses simultaneous band scores (SimBaS). However, at the end of the day, we have not added it to the paper because (in our understanding as non-experts in Bayesian statistics) SimBaS (as implemented in Meyer et al. (2015)) is not fully compatible with our model setting: The model, estimation pipeline, and calibration assumptions of SimBaS all target function-on-function surfaces $\beta(v, t)$ with non-degenerate functional predictors and multiple curves. Forcing the publicly available code to match our setting requires adding jitter or pseudo-replication that changes the estimand and invalidates error control. Conceptually, the SimBaS idea may of course be generalized, but that is not what the available code does. More

[1] We used the code from `https://github.com/MorrisStatLab/BayesFMM_Function_on_Function` following Zemplenyi et al. (2021).

specifically, the main issue is this: As we have understood, SimBaS is defined by inverting joint credible bands over the 2-D coefficient surface $\beta(v,t)$ for the model $y_{ic}(t) = \int_{v \in V} x_{ic}(v)\beta(v,t)dv + U_i(t) + E_{ic}(t)$ after transforming both $y_{ic}(t)$ and $x_{ic}(v)$ to basis space and standardizing the posterior draws; it then computes $P_{\text{SimBaS}}(v,t)$ using the global maximum over the (v,t) domain (equations (7)–(8) in Zemplenyi et al. (2021)). This controls experiment-wise error across $V \times T$ locations. If we want to match this model with ours, we need to assume that $x_{ic}(v) \equiv 1$. The problem then collapses to a function-on-scalar coefficient $b(t) := \int_{v \in V} \beta(v,t)dv$, for which the SimBaS surface machinery does not seem to be appropriate. Assuming $x_{ic}(v) \equiv 1$ in particular creates problems in terms of implementation. The WFFR procedure that SimBaS relies on always applies a discrete wavelet transform (DWT) to each row of $\mathbf{X} = (x_{ic}(v))_{i,c}$ and then performs wavelet-space PCA before fitting the model. With $x_{ic}(v) \equiv 1$, the DWT of $\mathbf{X}$ is degenerate, the standardization step results in dividing by zero, but the code expects wavelet-spec fields and dimensions that don't exist in this degenerate case. In summary, (as non-experts in Bayesian statistics) we believe that as currently implemented, SimBaS is not compatible with our model setting and thus cannot be regarded as an off-the-shelf benchmark. For this reason, we have not considered it in the revision.

3. *Stochastic volatility (SV) is widely important in the types of applications that the authors mention. Yet the authors 1) claim that extensions to SV would be challenging, 2) elect not to consider any simulation designs with SV to examine robustness, and 3) claim that SV can be handled with log-transformations (and then using the same mean models as proposed). Perhaps 1) is true, but 2) is important based on the claims in the paper and the relevance for this journal. Finally, 3) is not correct for the most widely-used SV models.*

   With stochastic volatility, we guess you mean time-varying (short-run and long-run) error variance.

   ad 2) As a robustness check, we have included a design with time-varying error variance in the revised simulation study. More specifically, we assume that for each i, the idiosyncratic error process $\{\varepsilon_{it} : 1 \le t \le T\}$ follows an AR(1) model with a time-varying (short- and long-run) variance. The details can be found in Section A.2.4 of the supplement.

   ad 3) We fully agree that log-transformations only work in special cases. We mentioned them in our reply letter to your previous referee report only to point out that at least in some special cases, our methods can easily accommodate time-varying error variances.

Reply to referee 2

Thank you very much for the careful reading of our manuscript. We have addressed your remaining question in the revision. Please see our reply below.

1. *Thank you very much for your very careful revision. I have just one remaining "cosmetic" question: can you explain why in the third row of your figures such as Figure 1, page 27, we see several intervals one on top of the other? In other words is there a meaning of the y-axis of this lower plots, what is its scale? Thank you in advance for adding this information (which I might have missed elsewhere?).*

   The third row of our figures (such as Figure 1) shows the intervals that are contained in the sets $\mathcal{S}^{[i,j]}(\alpha)$. To make these intervals all visible, we plot them one on top of the other. However, the order in which we plot them is completely arbitrary, i.e., their vertical positioning does not contain any information. We have clarified this in the figure caption.

References

Knoll, K., Schularick, M. and Steger, T. (2017). No price like home: global house prices, 1870-2012. *American Economic Review*, **107** 331–53.

Meyer, M. J., Coull, B. A., Versace, F., Cinciripini, P. and Morris, J. S. (2015). Bayesian function-on-function regression for multilevel functional data. *Biometrics*, **71** 563–574.

Zemplenyi, M., Meyer, M. J., Cardenas, A., Hivert, M.-F., Rifas-Shiman, S. L., Gibson, H., Kloog, I., Schwartz, J., Oken, E., DeMeo, D. L. et al. (2021). Function-on-function regression for the identification of epigenetic regions exhibiting windows of susceptibility to environmental exposures. *The Annals of Applied Statistics*, **15** 1366.